



Objective

- **Business Question:** *Which customers are likely to churn?*
- **Clarifications:**
 - *Churn* → customer leaving the service.
 - *Yes* → customer churned.
 - *No* → customer retained.
- **Importance:**
 - Helps the business prioritize retention strategies.
 - False positives → wasted retention effort.
 - False negatives → missed opportunity, customer lost.



Data Exploration (Phases 2–4)

- **Load Dataset**
 - Import libraries, load CSV, inspect dimensions, columns, data types, first/last records.
- **Understand Features**
 - Classify into numerical, categorical, identifier, target.
 - Interpret meaning of each feature.
- **Target Analysis**
 - Count churned vs retained customers.
 - Check percentages and class imbalance.
- **Data Quality Checks**
 - Missing values, duplicates, incorrect types, suspicious/inconsistent/impossible values, leakage.
 - Decide what to remove (e.g., customer ID, irrelevant/problematic features).
- **Exploratory Data Analysis (EDA)**
 - Univariate distributions (numerical, categorical, churn).
 - Churn vs features (tenure, monthly charges, contract type, payment method).
 - Extract business insights (associations, not causation).